

**Birla Institute of Technology & Science, Pilani**  
**Work-Integrated Learning Programmes Division**  
**Mid-Semester Test**  
**(EC-2 Regular)**

Course No.: SSZG527  
Duration : 2 Hours

Course Title: Cloud Computing  
Date of Exam: 15/03/2026 AN

Weightage: 30%

Note:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Answers should be specific and crisp including justification where required.
4. Assumptions made if any, should be stated clearly at the beginning of your answer.

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|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Q1 | Pick any ONE Virtualization technique of your choice, among the full virtualization with VMs or software virtualization using containers. State its TWO primary demerits in Cloud Computing with a very brief justification, and how can they be mitigated?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5<br>(3+2)   |
| Q2 | Write a Dockerfile to create an image for a web application that requires a specific version of Node.js (14.16.1) and a custom dependency (express 4.16.2). The image should be based on the official Ubuntu 22.04 image, have a working directory at /app, and copy the package.json file into the working directory. After installing the dependencies, copy the remaining application code into the working directory and expose port 3232. You need not write any docker commands to build the image or to run the container.<br><b>Output: A Dockerfile that generates the desired image when built.</b>                                                                                                                                                                                                                                                       | 5            |
| Q3 | A multi-specialty hospital is planning to modernize its Hospital Information System (HIS). The management is evaluating two alternative cloud-based architectures: <ul style="list-style-type: none"> <li>• <b>Architecture A:</b> A system primarily built using <b>SaaS solutions</b> for electronic health records, billing, and appointment scheduling, with minimal customization.</li> <li>• <b>Architecture B:</b> A system that combines <b>PaaS-based custom application development</b> with <b>IaaS-hosted virtual machines</b> for legacy clinical and diagnostic applications.</li> </ul> <p>a) Compare Architecture A and Architecture B in terms of <b>scalability, customization, and operational control</b>.</p> <p>b) Identify scenarios in which Architecture A would be preferred over Architecture B, and vice versa, with justification.</p> | 5<br>(3+2)   |
| Q4 | A global retailer subscribes to a cloud-based ERP system accessed via a browser for inventory management. At the same time, its IT team uses Azure virtual machines to host custom applications. Identify the service models for these TWO given situations and the deployment model overall. Justify your answer. Without justification no marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6<br>(2+2+2) |
| Q5 | What role do Docker volumes play for container restarts? How do they impact the overall portability and efficiency when multiple Docker containers are deployed in a production environment?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 (2+2)      |
| Q6 | A company may expand private cloud capacity by <b>20 units</b> , but expansion costs <b>Rs.1,800 (fixed)</b> . <ul style="list-style-type: none"> <li>• Private unit cost even after expansion remains Rs.100</li> <li>• Current private capacity = 40 units</li> <li>• Public cloud cost = Rs.240 per unit</li> <li>• Budget = Rs.10,000 (includes expansion if chosen)</li> </ul> <p><b>Question:</b><br/>Should the company expand private capacity to maximize total demand served? Justify by showing appropriate calculations.</p>                                                                                                                                                                                                                                                                                                                            | 5            |